Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2014 – Paper 3 – Solutions**

June 2014 – Question 1 – Solution

(a) The binary operation $*$ is defined on a set $(e, a, b, c, d, f\}$ as shown in the table below.

For example, $a * b = d$.

$*$	$\boldsymbol{e}$	$\boldsymbol{a}$	$\boldsymbol{b}$	$\boldsymbol{c}$	$\boldsymbol{d}$	$\boldsymbol{f}$
$\boldsymbol{e}$	e	a	b	c	d	f
$\boldsymbol{a}$	a	e	d	f	b	c
$\boldsymbol{b}$	b	f	e	d	c	a
$\boldsymbol{c}$	c	d	f	e	a	b
$\boldsymbol{d}$	d	c	a	b	f	e
$\boldsymbol{f}$	f	b	c	a	e	d

(i) State, giving a reason, if $*$ is commutative. [2]

For an operation to be commutative, $x * y = y * x$ for all elements.

From the table,

$a * b = d$

$b * a = f$

Since $d \neq f$, then $a * b \neq b * a$

$\therefore *$ is not commutative.

(ii) Name the identity element for the operation $*$. [1]

An identity element e satisfies $x * e = e * x = x$ for all x.

$*$	$\boldsymbol{e}$	$\boldsymbol{a}$	$\boldsymbol{b}$	$\boldsymbol{c}$	$\boldsymbol{d}$	$\boldsymbol{f}$
$\boldsymbol{e}$	e	a	b	c	d	f
$\boldsymbol{a}$	a	e	d	f	b	c
$\boldsymbol{b}$	b	f	e	d	c	a
$\boldsymbol{c}$	c	d	f	e	a	b
$\boldsymbol{d}$	d	c	a	b	f	e
$\boldsymbol{f}$	f	b	c	a	e	d

From the table, e is the identity element.

(iii) Determine the inverse of

(a) d [1]

Since $d * f = f * d = e$, then the inverse of d is $d^{-1} = f$.

(b) c [1]

Since $c * c = e$, then c is self-inverse.

$\therefore c^{-1} = c$

(b) Let α, β and γ be the roots of the equation $2x^3 + 4x^2 + 3x - 1 = 0$.

(i) Calculate EACH of the following:

(a) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$ [5]

$2x^3 + 4x^2 + 3x - 1 = 0$

which is in the form $ax^3 + bx^2 + cx + d = 0$,

where $a = 2, b = 4, c = 3$ and $d = -1$.

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$= -\frac{4}{2}$$

$$= -2$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a}$$

$$= \frac{3}{2}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

$$= \frac{-(-1)}{2}$$

$$= \frac{1}{2}$$

Now,

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{\beta^2\gamma^2 + \alpha^2\gamma^2 + \alpha^2\beta^2}{\alpha^2\beta^2\gamma^2}$$

$$= \frac{(\beta\gamma)^2 + (\alpha\gamma)^2 + (\alpha\beta)^2}{(\alpha\beta\gamma)^2}$$

Let us first represent the numerator using:

$$(\alpha\beta + \beta\gamma + \alpha\gamma)^2 = (\alpha\beta + \beta\gamma + \alpha\gamma)(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$= (\alpha\beta)^2 + \alpha\beta^2\gamma + \alpha^2\beta\gamma + \alpha\beta^2\gamma + (\beta\gamma)^2 + \alpha\beta\gamma^2 + \alpha^2\beta\gamma + \alpha\beta\gamma^2 + (\alpha\gamma)^2$$

$$= (\alpha\beta)^2 + (\beta\gamma)^2 + (\alpha\gamma)^2 + 2\alpha^2\beta\gamma + 2\alpha\beta^2\gamma + 2\alpha\beta\gamma^2$$

$$= (\alpha\beta)^2 + (\beta\gamma)^2 + (\alpha\gamma)^2 + 2\alpha\beta\gamma(\alpha + \beta + \gamma)$$

Rearranging this equation gives:

$$(\alpha\beta)^2 + (\beta\gamma)^2 + (\alpha\gamma)^2 = (\alpha\beta + \beta\gamma + \alpha\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$$

So, we have,

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{(\beta\gamma)^2 + (\alpha\gamma)^2 + (\alpha\beta)^2}{(\alpha\beta\gamma)^2}$$

$$= \frac{(\alpha\beta + \beta\gamma + \alpha\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)}{(\alpha\beta\gamma)^2}$$

$$= \frac{\left(\frac{3}{2}\right)^2 - 2\left(\frac{1}{2}\right)(-2)}{\left(\frac{1}{2}\right)^2}$$

$$= \frac{\frac{9}{4} + 2}{\frac{1}{4}}$$

$$= \frac{\left(\frac{17}{4}\right)}{\left(\frac{1}{4}\right)}$$

$$= \frac{17}{4} \div \frac{1}{4}$$

$$= \frac{17}{4} \times \frac{4}{1}$$

$$= 17$$

$$\therefore \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = 17$$

(b) $\frac{1}{\alpha^2\beta^2}+\frac{1}{\beta^2\gamma^2}+\frac{1}{\gamma^2\alpha^2}$ [2]

$$\frac{1}{\alpha^2\beta^2}+\frac{1}{\beta^2\gamma^2}+\frac{1}{\gamma^2\alpha^2}=\frac{\gamma^2+\alpha^2+\beta^2}{\alpha^2\beta^2\gamma^2}$$

Let us first represent the numerator using:

$$(\alpha+\beta+\gamma)^2=(\alpha+\beta+\gamma)(\alpha+\beta+\gamma)$$

$$=\alpha^2+\alpha\beta+\alpha\gamma+\alpha\beta+\beta^2+\beta\gamma+\alpha\gamma+\beta\gamma+\gamma^2$$

$$=\alpha^2+\beta^2+\gamma^2+2\alpha\beta+2\beta\gamma+2\alpha\gamma$$

$$=\alpha^2+\beta^2+\gamma^2+2(\alpha\beta+\beta\gamma+\alpha\gamma)$$

Rearranging this equation gives:

$$\alpha^2+\beta^2+\gamma^2=(\alpha+\beta+\gamma)^2-2(\alpha\beta+\beta\gamma+\alpha\gamma)$$

So, we have,

$$\frac{1}{\alpha^2\beta^2}+\frac{1}{\beta^2\gamma^2}+\frac{1}{\gamma^2\alpha^2}=\frac{\gamma^2+\alpha^2+\beta^2}{\alpha^2\beta^2\gamma^2}$$

$$=\frac{(\alpha+\beta+\gamma)^2-2(\alpha\beta+\beta\gamma+\alpha\gamma)}{(\alpha\beta\gamma)^2}$$

$$=\frac{(-2)^2-2\left(\frac{3}{2}\right)}{\left(\frac{1}{2}\right)^2}$$

$$=\frac{4-3}{\left(\frac{1}{4}\right)}$$

$$=\frac{1}{\left(\frac{1}{4}\right)}$$

$$=1\div\frac{1}{4}$$

$$=1\times\frac{4}{1}$$

$$=4$$

(ii) Hence, or otherwise, find the equation whose roots are $\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}$. [2]

$$\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}=17$$

$$\frac{1}{\alpha^2\beta^2}+\frac{1}{\beta^2\gamma^2}+\frac{1}{\gamma^2\alpha^2}=4$$

Consider,

$$\left(\frac{1}{\alpha^2}\right)\left(\frac{1}{\alpha^2}\right)\left(\frac{1}{\gamma^2}\right)=\frac{1}{\alpha^2\beta^2\gamma^2}$$

$$=\frac{1}{(\alpha\beta\gamma)^2}$$

$$=\frac{1}{\left(\frac{1}{2}\right)^2}$$

$$=\frac{1}{\left(\frac{1}{4}\right)}$$

$$=1\div\frac{1}{4}$$

$$=1\times\frac{4}{1}$$

$$=4$$

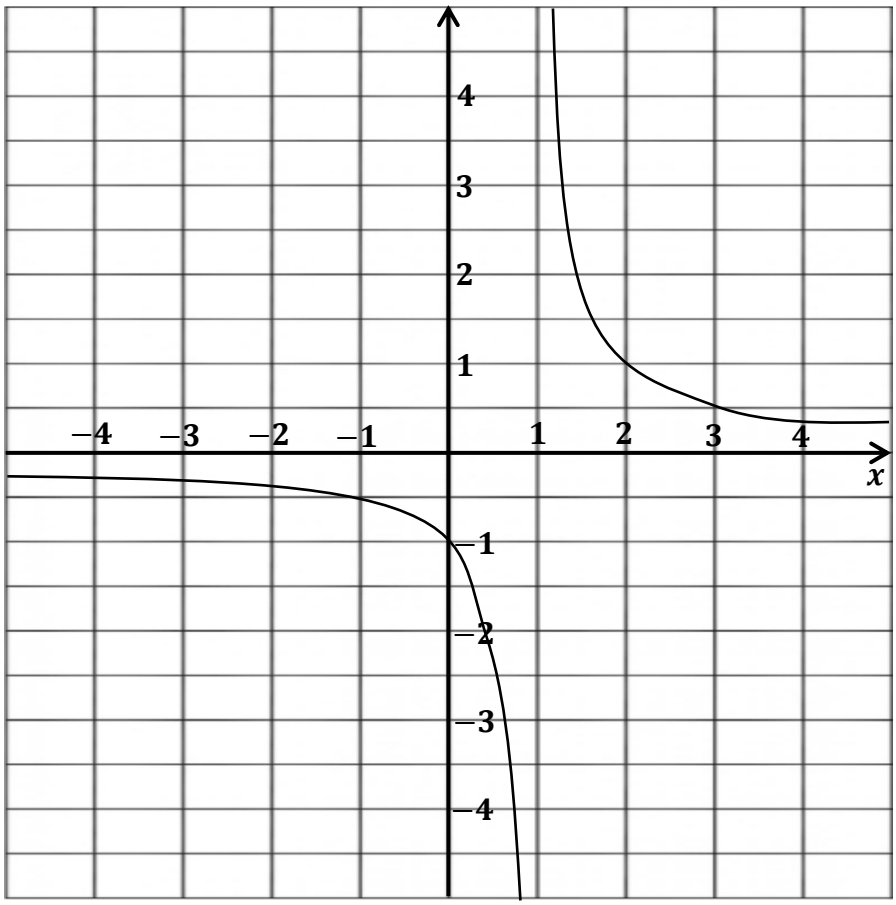
The new equation is:

$$x^3-(sum\ of\ roots)x^2+(sum\ of\ the\ product\ of\ two\ of\ the\ roots\ at\ a\ time)x-(product\ of\ the\ roots)=0$$

$$x^3-17x^2+4x-4=0$$

(c) **An answer sheet is provided for this question.**

The diagram below shows the graph of the function $g(x) = \frac{1}{x-1}$.



(i) **On the answer sheet provided**, sketch the graphs of $|g(x)|$ and $f(x) = x - 1$, showing clearly the intercepts and the asymptotes. [5]

$g(x) = \frac{1}{x-1}$

The graph of $|g(x)| = \left|\frac{1}{x-1}\right|$ is always above the x-axis.

For $x > 1$, the graph is the same as $g(x)$.

For $x < 1$, the part of $g(x)$ below the x-axis is reflected above the x-axis.

There are no x-intercepts.

For $g(x)$, the coordinates of the y-intercept is $(0, -1)$.

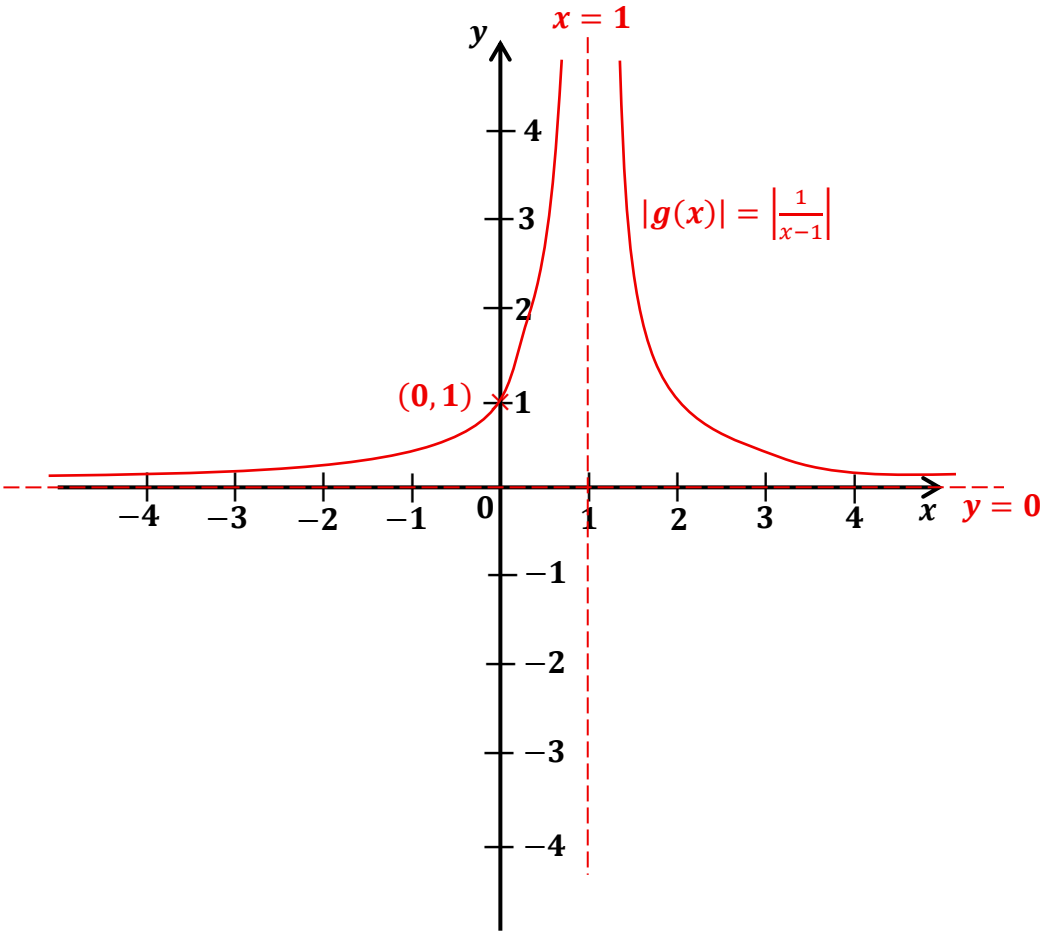
For $|g(x)|$, the coordinates of the y-intercept will be $(0, 1)$.

The vertical asymtote is $x = 1$.

The horizontal asymptote is $y = 0$.

The graph is shown below:

Answer Sheet for Question 1(c)



(ii) Hence, or otherwise, obtain the value of x such that $f(x) = g(x)$. [1]

We have $f(x) = x - 1$ and $g(x) = \frac{1}{x-1}$.

Now,

$$f(x) = g(x)$$

$$x - 1 = \frac{1}{x-1}$$

$$(x-1)^2 = 1$$

$$x^2 - 2x + 1 = 1$$

$$x^2 - 2x = 0$$

$$x(x-2) = 0$$

Either $x = 0$ or $x - 2 = 0$

$x = 2$

$\therefore x = 0$ and $x = 2$

Total: 20 marks

June 2014 – Question 2 – Solution

(a) $P(1,3,2)$, $Q(-1,2,3)$ and $R(1,3,5)$ are the vertices of a triangle.

(i) Find the displacement vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$. [4]

We are given that $P(1,3,2)$, $Q(-1,2,3)$ and $R(1,3,5)$.

So, we have the following position vectors:

$$\overrightarrow{OP} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} \quad , \quad \overrightarrow{OQ} = \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OR} = \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix}$$

Using the triangle law,

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$$

$$= \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$$

Using the triangle law,

$$\overrightarrow{PR} = \overrightarrow{OR} - \overrightarrow{OP}$$

$$= \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$$

$\therefore$ The displacement vectors are $\overrightarrow{PQ} = \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$ and $\overrightarrow{PR} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$.

(ii) Hence, determine

(a) $|\overrightarrow{PQ}|$ and $|\overrightarrow{PR}|$ [3]

$$\overrightarrow{PQ} = \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$$

$$|\overrightarrow{PQ}| = \sqrt{(-2)^2 + (-1)^2 + (1)^2}$$

$$= \sqrt{4 + 1 + 1}$$

$$= \sqrt{6} \text{ units}$$

$$\overrightarrow{PR} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$$

$$|\overrightarrow{PR}| = \sqrt{(0)^2 + (0)^2 + (3)^2}$$

$$= \sqrt{0 + 0 + 9}$$

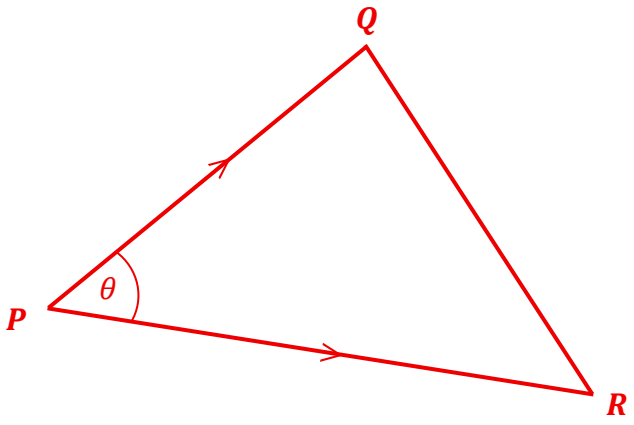
$$= \sqrt{9}$$

$$= 3 \text{ units}$$

$\therefore |\overrightarrow{PQ}| = \sqrt{6}$ units and $|\overrightarrow{PR}| = 3$ units.

(b) the cosine of the acute angle between $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ [4]

Consider the diagram below.



$$\overrightarrow{PQ} \cdot \overrightarrow{PR} = \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$$

$$= (-2)(0) + (-1)(0) + (1)(3)$$

$$= 0 + 0 + 3$$

$$= 3$$

Now,

$$\cos\theta = \frac{\overrightarrow{PQ} \cdot \overrightarrow{PR}}{|\overrightarrow{PQ}||\overrightarrow{PR}|}$$

$$\cos\theta = \frac{3}{(\sqrt{6})(3)}$$

$$\cos\theta = \frac{1}{\sqrt{6}}$$

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{6}}\right)$$

$$\theta = 65.9° \qquad \text{(to 1 decimal place)}$$

(c) the area of triangle PQR [4]

Area of triangle $PQR = \frac{1}{2}|\overrightarrow{PQ}||\overrightarrow{PR}|\cos\theta$

$= \frac{1}{2}(\sqrt{6})(3)\left(\frac{1}{\sqrt{6}}\right)$

$= \frac{3}{2}\ units^2$

$\therefore$ The area of triangle PQR is $\frac{3}{2}\ units^2$.

(b) Given that $\frac{\pi}{12} = \frac{\pi}{3} - \frac{\pi}{4}$, show **without the use of a calculator**, that the EXACT value of $\tan\frac{\pi}{12}$ is $2 - \sqrt{3}$. [5]

$$\tan\left(\frac{\pi}{12}\right) = \tan\left(\frac{\pi}{3} - \frac{\pi}{4}\right)$$

$$= \frac{\tan\left(\frac{\pi}{3}\right) - \tan\left(\frac{\pi}{4}\right)}{1 + \tan\left(\frac{\pi}{3}\right)\tan\left(\frac{\pi}{4}\right)}$$

$$= \frac{(\sqrt{3}) - (1)}{1 + (\sqrt{3})(1)}$$

$$= \frac{\sqrt{3} - 1}{1 + \sqrt{3}}$$

$$= \frac{\sqrt{3} - 1}{1 + \sqrt{3}} \times \frac{1 - \sqrt{3}}{1 - \sqrt{3}}$$

$$= \frac{\sqrt{3} - 3 - 1 + \sqrt{3}}{(1)^2 - (\sqrt{3})^2}$$

$$= \frac{2\sqrt{3} - 4}{1 - 3}$$

$$= \frac{2\sqrt{3} - 4}{-2}$$

$$= \frac{2(\sqrt{3} - 2)}{2(-1)}$$

$$= \frac{\sqrt{3} - 2}{-1}$$

$$= -\sqrt{3} + 2$$

$$= 2 - \sqrt{3}$$

$$\therefore \tan\left(\frac{\pi}{12}\right) = 2 - \sqrt{3}$$

Q.E.D.

Total: 20 marks

June 2014 – Question 3 – Solution

(a) Evaluate $\lim_{x \to 0} \frac{\sin 8x}{2x}$. [4]

$$\lim_{x \to 0} \frac{\sin 8x}{2x} = \frac{1}{2} \lim_{x \to 0} \frac{\sin 8x}{x}$$

$$= \frac{1}{2}(8) \lim_{x \to 0} \frac{\sin 8x}{8x}$$

$$= 4 \lim_{x \to 0} \frac{\sin 8x}{8x}$$

$$= 4(1)$$

$$= 4$$

$$\therefore \lim_{x \to 0} \frac{\sin 8x}{2x} = 4$$

(b) The equation of a curve is given by

$$y = x^3 + x^2 + 2$$

(i) Determine the coordinates of the points on the curve where the gradient is 1. [6]

$$y = x^3 + x^2 + 2$$

$$\frac{dy}{dx} = 3x^2 + 2x$$

When $\frac{dy}{dx} = 1$,

$$1 = 3x^2 + 2x$$

$$0 = 3x^2 + 2x - 1$$

$$0 = (3x - 1)(x + 1)$$

Either $3x - 1 = 0$ or $x + 1 = 0$

$3x = 1$ $x = -1$

$x = \frac{1}{3}$

When $x = \frac{1}{3}$,

$y = \left(\frac{1}{3}\right)^3 + \left(\frac{1}{3}\right)^2 + 2$

$= \frac{1}{27} + \frac{1}{9} + 2$

$= \frac{58}{27}$

When $x = -1$,

$y = (-1)^3 + (-1)^2 + 2$

$= -1 + 1 + 2$

$= 2$

$\therefore$ The coordinates of the points on the curve are $\left(\frac{1}{3}, \frac{58}{27}\right)$ and $(-1, 2)$.

(ii) Determine the equation of the normal which intersect the curve at $(-1, 2)$. [4]

$\frac{dy}{dx} = 3x^2 + 2x$

When $x = -1$,

$\frac{dy}{dx} = 3(-1)^2 + 2(-1)$

$= 3(1) - 2$

$= 3 - 2$

$= 1$

Gradient of tangent $= 1$

Gradient of normal $= -1$

Substituting $m = -1$ and $(-1, 2)$ into $y - y_1 = m(x - x_1)$ gives:

$y - 2 = -1\big(x - (-1)\big)$

$y - 2 = -1(x + 1)$

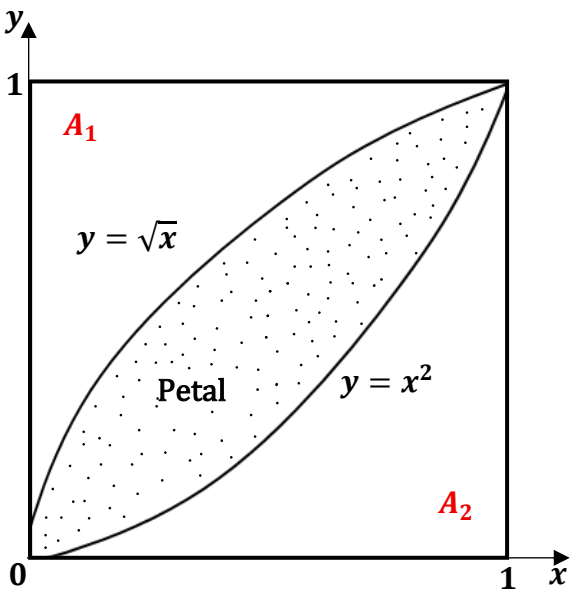
$y - 2 = -x - 1$

$y = -x - 1 + 2$

$y = -x + 1$

$\therefore$ The equation of the normal is: $y = -x + 1$

(c) The diagram below (**not drawn to scale**) shows the design of a petal drawn on a square tile of length 1 metre.



The design may be modelled by the finite region enclosed by the curves $y = \sqrt{x}$ and $y = x^2$ where x and y are lengths measured in metres.

Calculate the area of the petal. [6]

Curve 1:

$y = \sqrt{x}$

$y^2 = x$

$x = y^2$

Curve 2:

$y = x^2$

$x^2 = y$

$x = \sqrt{y}$

Area of A_1 = Area of Curve 2 − Area of Curve 1

$$= \int_0^1 \sqrt{y}\, dy - \int_0^1 y^2\, dy$$

$$= \int_0^1 (\sqrt{y} - y^2)\, dy$$

$$= \int_0^1 \left(y^{\frac{1}{2}} - y^2\right) dy$$

$$= \left[\frac{y^{\frac{3}{2}}}{\frac{3}{2}} - \frac{y^3}{3}\right]_0^1$$

$$= \left[\frac{2}{3}y^{\frac{3}{2}} - \frac{y^3}{3}\right]_0^1$$

$$= \left[\frac{2}{3}(1)^{\frac{3}{2}} - \frac{(1)^3}{3}\right] - \left[\frac{2}{3}(0)^{\frac{3}{2}} - \frac{(0)^3}{3}\right]$$

$$= \left(\frac{2}{3} - \frac{1}{3}\right) - 0$$

$$= \frac{1}{3}\, m^2$$

<u>Curve 1:</u> $y = \sqrt{x}$ and <u>Curve 2:</u> $y = x^2$

Area of A_2 = Area of Curve 1 − Area of Curve 2

$$= \int_0^1 \sqrt{x}\, dx - \int_0^1 x^2\, dx$$

$$= \int_0^1 (\sqrt{x} - x^2)\, dx$$

$$= \int_0^1 \left(x^{\frac{1}{2}} - x^2\right) dx$$

$$= \left[\frac{2}{3}x^{\frac{3}{2}} - \frac{x^3}{3}\right]_0^1$$

$$= \left[\frac{2}{3}(1)^{\frac{3}{2}} - \frac{(1)^3}{3}\right] - \left[\frac{2}{3}(0)^{\frac{3}{2}} - \frac{(0)^3}{3}\right]$$

$$= \left(\frac{2}{3} - \frac{1}{3}\right) - 0$$

$$= \frac{1}{3}\, m^2$$

Now,

Area of square tile $= s \times s$

$= 1 \times 1$

$= 1\ m^2$

So, we have,

Area of petal = Area of square tile − (Area of A_1 + Area of A_2)

$= 1 - \left(\frac{1}{3} + \frac{1}{3}\right)$

$= 1 - \frac{2}{3}$

$= \frac{1}{3}\ m^2$

∴ The area of the petal is $\frac{1}{3}\ m^2$.

Total: 20 marks